Social Media Marketing Performance Analysis

Zoe Nguyen

3/19/2026

Project Domain: Marketing Data Analytics

1. Tool

- **SQL:** Data querying and aggregation.
- **Python** (Pandas & Seaborn)**:** Data cleaning and exploratory data analysis (EDA).
- **Tableau:** Interactive dashboarding and data visualization.

2. Executive Summary

Project Overview

Analyzed a comprehensive dataset of **5,600+** social media posts from 2024 using SQL, Python, and Tableau to optimize digital marketing ROI for a global IT firm.

Key Insights

- Platform: Instagram leads at **15.85%** ER, LinkedIn underperforms at 14.30%
- Content: Customer Story content dominates at **20.22% ER across all formats**
- Timing: Monday **6 PM** is the peak slot at 18.00% ER
- Distribution: Organic delivers **4x more** views than Sponsored at equal ER

View the Interactive Dashboard on [Tableau Public]

3. Platform Performance

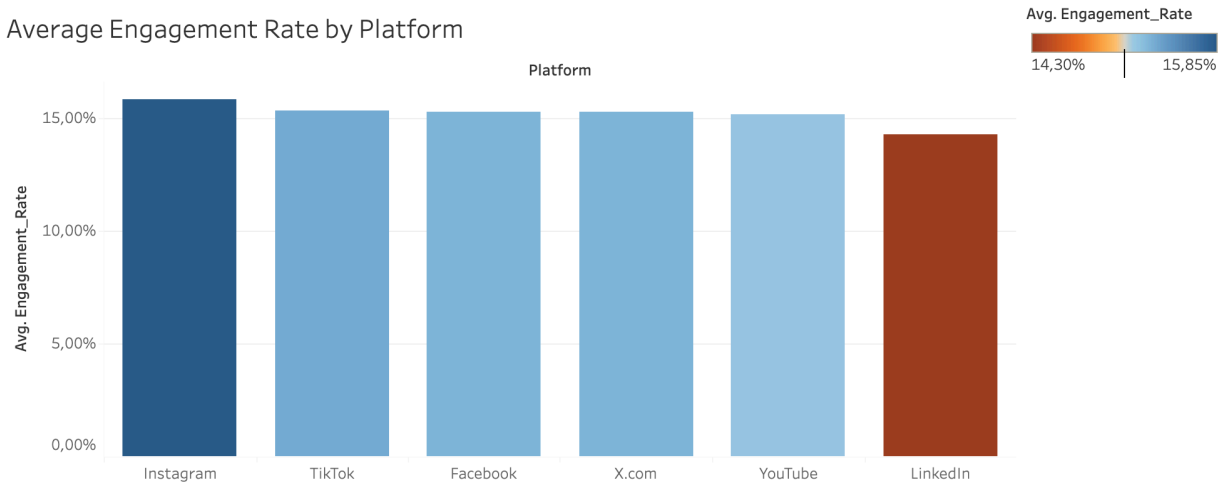

Figure 1: Average Engagement Rate by Platform (2024)

- **Insight:** Instagram leads all platforms with a 15.85% average ER — the only

platform reaching the top of the color scale.

- **Evidence:** LinkedIn sits at the opposite end at ~14.30%, making the gap between best and worst nearly 1.5 percentage points across 5,600+ posts.
- **Implication:** Prioritize Instagram for brand awareness campaigns. Reposition LinkedIn content toward B2B lead generation rather than chasing engagement metrics; it structurally cannot win.

4. Content Formula

Content Formula

	Content_Category				
Post_..	Customer Story	Educational	Entertainment	Event / Webinar	Product Promotion
Image	19,63%	19,99%	7,78%	14,15%	11,54%
PDF		20,60%	9,00%	14,00%	17,22%
Text	19,62%	19,96%	7,77%	13,66%	11,19%
Article	20,07%	19,78%	7,33%		11,12%
Live Stream	20,11%	19,93%		13,91%	11,56%
Carousel	20,22%	19,25%	7,46%		12,96%
Video	20,00%	19,81%	7,46%	15,68%	11,60%

Figure 2: Engagement Rate by Content Type and Category

- **Insight:** Customer Story is the single strongest content category — regardless of post format.
- **Evidence:** The entire Customer Story column ranges from 19.62% to 20.22% ER, nearly 3x higher than Entertainment (7.33%–9.00%). This holds across Video, Carousel, Text, and Live Stream — confirming it's the *category*, not the format, driving performance.
- **Implication:** Systematically collect customer success stories and repurpose them across all formats. Stop over-investing in entertainment content.

5. Golden Hour

Golden Hour Heatmap

P..	DayOfWeek						
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
18	18,00%	15,12%	14,71%	15,14%	15,69%	16,43%	14,57%
16	15,15%	16,52%	16,24%	15,61%	14,21%	15,27%	15,87%
11	15,78%	15,33%	14,95%	15,83%	15,30%	16,06%	15,36%
13	14,19%	17,22%	15,77%	14,71%	17,52%	14,10%	15,67%
14	14,70%	17,12%	17,11%	15,08%	13,67%	16,62%	13,57%
17	16,05%	14,65%	15,92%	15,69%	16,02%	14,59%	15,40%
15	15,68%	16,11%	15,68%	14,88%	14,75%	15,24%	15,93%
12	14,93%	15,24%	15,37%	15,03%	15,44%	14,68%	16,24%
8	16,93%	14,70%	14,01%	15,79%	14,99%	15,16%	16,18%
10	15,37%	14,75%	14,38%	15,85%	14,50%	15,30%	15,49%
19	14,67%	14,30%	12,92%	15,88%	16,36%	16,60%	15,66%
9	13,80%	14,36%	16,27%	14,93%	16,14%	14,54%	15,49%

Figure 3: Heatmap of Engagement by Day and Hour

- **Insight:** Monday 6:00 PM is the single highest-performing time slot at 18.00% ER, outperforming the assumed "mid-week afternoon" window.
- **Evidence:** Secondary peaks cluster Tuesday–Wednesday 13h–14h (~17%). The worst slot is Wednesday 9:00 AM at 12.92% — a 5-point swing from peak to trough.
- **Implication:** Launch hero content on Monday evenings. Use Tuesday-Wednesday afternoons for supporting posts. Avoid Wednesday mornings entirely.

6. Organic vs Sponsored

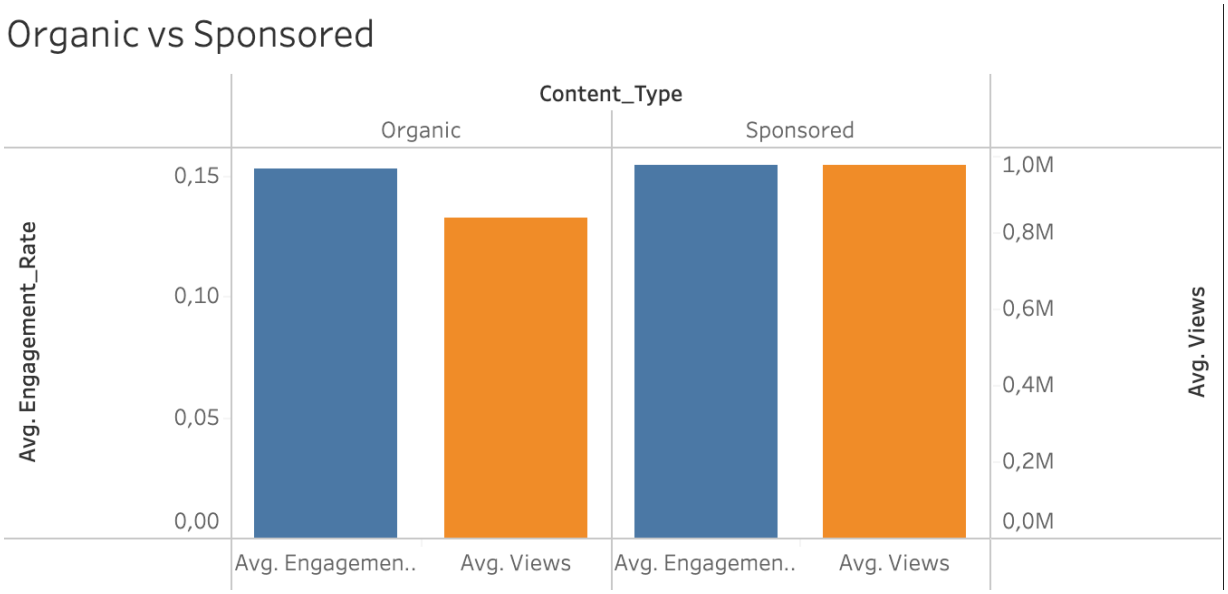

Figure 4: Organic vs Sponsored — Engagement Rate & Total Views

- **Insight:** Organic content delivers 4x more views than Sponsored — at the same engagement rate.
- **Evidence:** Organic: 3,877.4M views / 709 posts. Sponsored: 928.9M views / 147 posts. ER is nearly identical for both (~0.15), meaning paid spend is not buying better engagement quality.
- **Implication:** Shift budget from paid amplification into producing more Organic Customer Story content — the data shows it outperforms on both reach and authenticity.

7. Recommendations

- Double down on Customer Stories: Repurpose customer success content across all formats — it outperforms every other category by 3x.
- Automate the posting schedule: Deploy hero content Monday 18h, supporting content Tuesday–Wednesday 13h–14h.
- Rebalance paid vs organic: Redirect sponsored budget into producing more organic content — the data shows paid spend adds reach but not engagement quality.